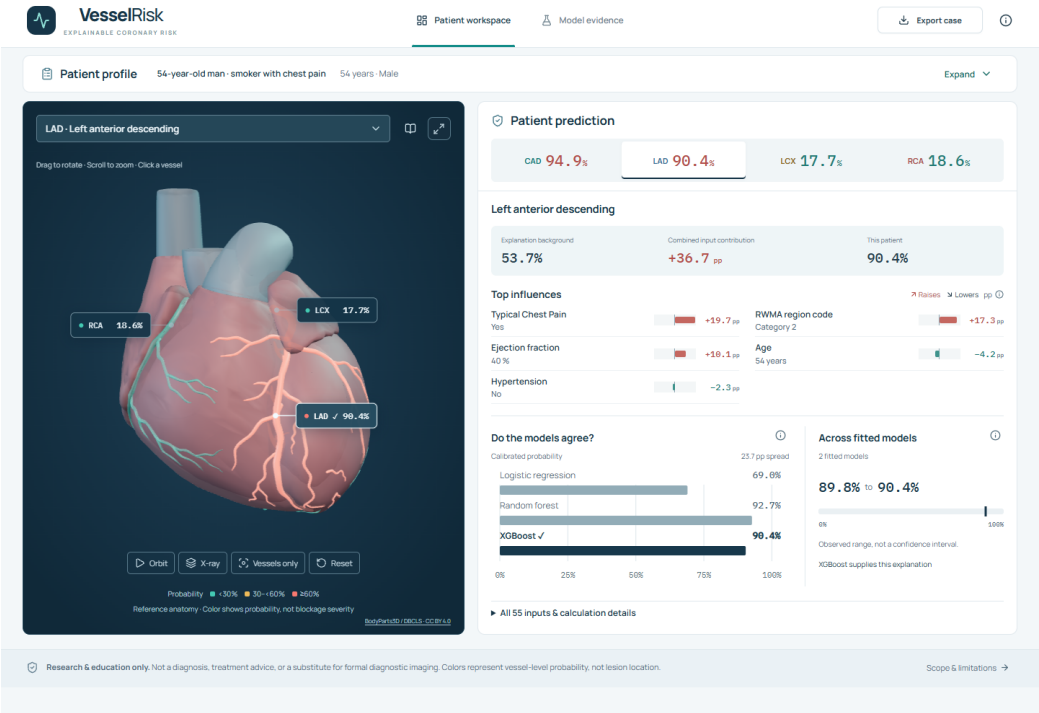
01 / PRODUCT & OBJECTIVE

Coronary risk, made visible.

Four calibrated clinical predictions, independently mapped to real coronary anatomy and explained by patient-level evidence.

VesselRisk connects **structured patient inputs** → **four trained classifiers** → **calibrated probabilities** → **interpretable 3D coronary anatomy**. Patient workspace presents the current profile, anatomy, signed contributions and model disagreement. Model evidence separately exposes the fixed training workflow, validation, calibration, failures and provenance; patient edits do not change those results.



Actual application screenshot: a held-out cohort case, not a mockup. Outputs are CAD 94.9%, LAD 90.4%, LCX 17.7% and RCA 18.6%. Selected demonstration cases are illustrative, not the evaluation sample.

Track A fit

Official criterion	Delivered evidence
Prediction · 30%	Four targets; nested repeated CV; model/calibration comparisons; honest per-target metrics
3D · 25%	63 licensed meshes; separate coronary territories; rotate, zoom, select, X-ray and vessel isolation
Interpretability · 20%	Exact-probability SHAP reconstruction, measured inputs, global ranks, model disagreement
Integration · 15% / technical · 10%	Validated API; synchronized outputs; local assets; pinned artifacts, hashes, tests and startup

02 / DATA & SCIENTIFIC PROTOCOL

Validation without shortcuts.

Target columns are rejected during fitting, preprocessing and inference.



Selection, preprocessing and calibration use outer training data only; test labels stay isolated.

Source workbook	Actual inspection
Populated worksheet	303 patients x 59 columns; 55 inputs. Empty Sheet1 excluded.
Quality audit	0 missing / duplicate rows; constant Exertional CP retained.
Positive / negative	CAD 216 / 87; LAD 177 / 126; LCX 119 / 184; RCA 114 / 189
Label discrepancy	Patient row 94: Cath Normal, LAD Stenotic. Source labels preserved.

Repeated nested stratified cross-validation

Outer: stratified 5-fold x 2 repeats, seed 2026; average two held-out probabilities per patient. **Inner:** three folds, six fixed candidates; maximize ROC-AUC - 0.25 x Brier. Select parameters, then family.

Family	Search / fixed configuration
Logistic regression	C = 0.1 or 1; balanced class weights; max_iter 2500
Random forest	min_samples_leaf = 3 or 8; 120 trees; max_features .7; balanced_subsample
XGBoost	max_depth = 2 or 3; 110 trees; rate .04; lambda 5; min_child_weight 4

Fold-local pipeline: numeric median imputation/scaling; categorical mode imputation/one-hot. No global selection, SMOTE or merging. Class weights are training-only; XGBoost uses regularization.

Calibration: predeclared sigmoid; three-fold OOF within outer train, then base-estimator refit (`ensemble=False`). Raw/isotonic: sensitivity only. Full-cohort artifacts use inner selection and refit.

Scope: headline metrics average two held-out models per patient; the workspace explains one fitted model. No external validation. Operating points appear separately (page 3).

03 / MEASURED PREDICTIVE PERFORMANCE

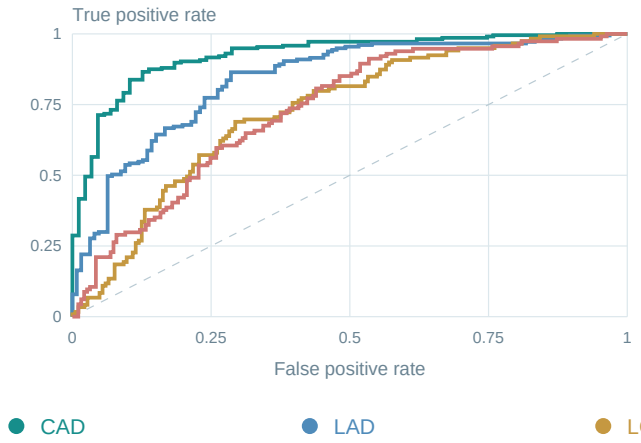
Strong CAD; harder vessel tasks.

Primary metrics use one repeat-averaged held-out probability per patient. Scores below are from shipped artifacts.

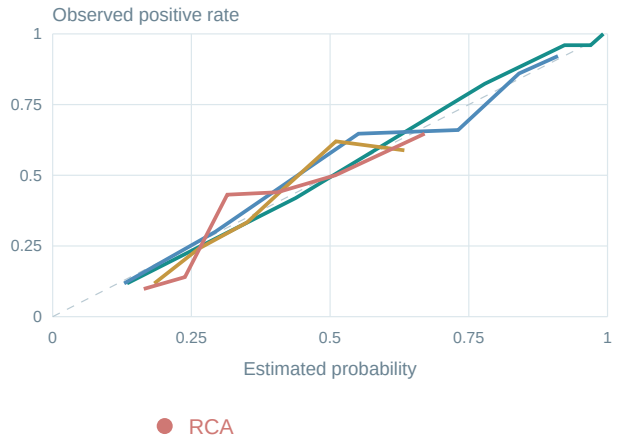
Target	AUC [95% CI]	Acc	Prec	Recall	F1	AP	Brier ↓
CAD	0.925 [0.891, 0.954]	0.868	0.896	0.921	0.909	0.967	0.094
LAD	0.839 [0.791, 0.886]	0.789	0.793	0.864	0.827	0.868	0.157
LCX	0.728 [0.670, 0.786]	0.673	0.625	0.420	0.503	0.575	0.206
RCA	0.732 [0.672, 0.790]	0.660	0.570	0.395	0.466	0.577	0.201

Intervals: 1,000 patient-stratified bootstrap resamples conditional on fitted folds. They omit retraining/selection uncertainty and are not external-validation intervals. Fold SD is descriptive because folds overlap. AP is average precision.

ROC / all four targets



Calibration / six quantile bins

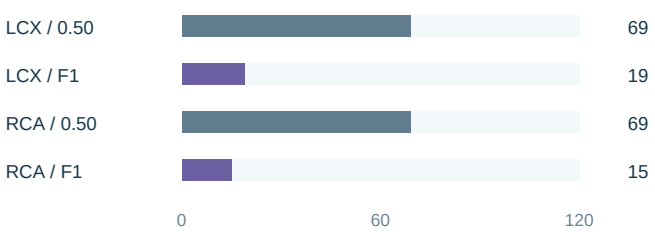


CAD LAD LCX RCA

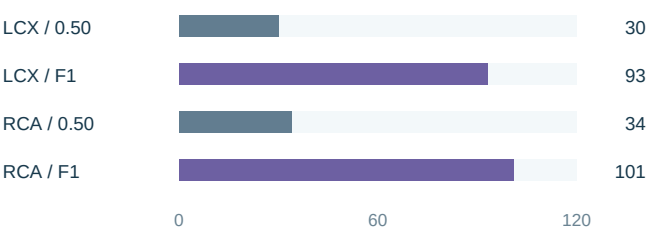
Probability quality: Brier error reduction versus training-fold prevalence: CAD 53.9%; LAD 35.2%; LCX 13.5%; RCA 14.2%. Model evidence adds group counts, descriptive Wilson intervals, conditional Brier intervals and separate-repeat scores. Intervals exclude refitting/selection uncertainty.

Training-selected research operating points

Missed positives (patients)



False positives (patients)



Gray: fixed 0.50; purple: training-selected F1. Same models/probabilities. LCX recall .420 to .840 (119 positives); RCA .395 to .868 (114 positives). Negative counts: 184 / 189. F1 cutoffs use inner OOF from outer train only; pool mean probability-minus-cutoff per patient. AUC/Brier stay unchanged. Accuracy falls .673 to .630 / .660 to .617. Research tradeoffs, not clinical cutoffs.

Bounded model expansions did not justify replacement: LCX/RCA AUC .731/.733 versus .728/.732, with paired gain intervals including zero. Reused splits are development evidence, not independent confirmation. Full trials: experiments/final_audit.

04 / INTERPRETABILITY & UNCERTAINTY

Explain the actual probability.

A contribution must explain the value the user sees, not an unrelated raw model margin.



Base + all signed contributions reconstruct the displayed calibrated probability.

Local SHAP: permutation explanations use the selected calibrated classifier and training-only background. Original categorical variables remain indivisible coalitions. Two cycles, 16/20 background patients, all 55 contributions. Reconstruction to 1e-6 checks additivity, not attribution precision.

Measured local explanation sensitivity

Target	Seed: top-5 overlap	Background: top-5 overlap	Max background change
CAD	85%	75%	10.7 pp
LAD	95%	75%	6.2 pp
LCX	95%	75%	3.9 pp
RCA	95%	75%	8.4 pp

Four fixed demo rows x four targets, using held-out models and training-only backgrounds. Compare seed, two vs eight cycles, and backgrounds at eight cycles with equal seed. Descriptive sensitivity, not population-wide stability.

Global SHAP uses six held-out patients x ten folds, background n=16 and one cycle. Fold rank correlations range .730-.775. Correlated inputs can share attribution. Full local vectors, reference variants and fingerprints: experiments/reliability/results.json.

Three complementary limits on confidence

Signal	Meaning and limitation
Candidate disagreement	Probability spread across three families exposes sensitivity to model choice. Not a confidence interval.
Refitted model range	Min/max of ten outer models; demo uses two excluding that patient. Descriptive; may omit the full-cohort estimate.
Profile representation	Standardized mixed-feature nearest-neighbor distance, cohort leave-one-out 95th percentile + single-feature range checks. Unvalidated heuristic.

Inspect failures, not only successes

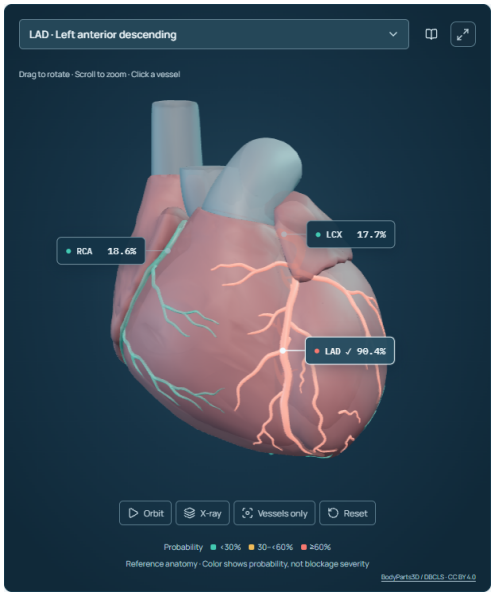
Model evidence exposes sex/age subgroups, eight largest held-out errors per target and fixed-logistic 5-fold ECG/laboratory-echo ablations. Small samples support descriptive checks; the original conflicting source labels are retained.

Boundary: editing inputs explores model response, not treatment effect. Independent CAD/vessel classifiers may disagree. Attribution does not establish causality or clinical validity.

05 / 3D & END-TO-END INTEGRATION

Real anatomy, honest mapping.

A memorable interaction with traceable structure identity, not invented lesion localization.



Actual interactive viewer. Heart-wall meshes, great vessels and independently selectable coronary branches are loaded locally; no asset network dependency during the demo.

Vessel mapping	Source geometry / identity
LAD	Anterior interventricular / anterior descending branches; trunk FJ2631
LCX	Circumflex branches of left coronary artery; FJ2649-FJ2654
RCA	Right coronary artery trunk and branches; trunk FJ2723

Asset: BodyParts3D 4.0, DBCLS, CC BY 4.0; pinned human-atlas derivative. 63 meshes, 37,444 triangles, 1,071,048 bytes. Selected, recentered and uniformly scaled; source names/IDs retained. Generic adult male anatomy, not a patient image. No RWMA-to-lesion mapping or geometry deformation.

Interaction: rotate, zoom, pan, select, orbit, X-ray, isolation, reset and expand. Surface-anchored labels show each artery probability and avoid overlap. A faint selected-vessel silhouette reveals occluded geometry. New predictions animate colors and briefly show changes from the previous prediction, not treatment effects. Orbit starts only on first opening. Pixel ratio is capped at 1.5; software WebGL checks do not establish universal FPS.

System path: Next.js clinical form → validated FastAPI request → calibrated CAD/LAD/LCX/RCA models → SHAP/uncertainty/OOD → probability, anatomy and explanation updates. Edits mark previous results stale and block exports; failures preserve results for retry. Input origins distinguish cohort records, defaults, user entries, imports and derived BMI; exports snapshot origins with exact values.

06 / REPRODUCIBILITY, USAGE & LIMITS

Ready to run and review.

The software, model weights and evidence run locally without cloud credentials or inference APIs.

Run the complete system

Python 3.13 and Node 24 recommended. First run: `python scripts/start.py --setup`. Subsequent launches: `python scripts/start.py`. Open **http://127.0.0.1:3000**. Select a held-out case → Predict → select vessels → inspect explanations → Model evidence. Optional Docker Compose packaging is supplied; Docker execution was not verified on this host.

Reproduce / inspect	Command or artifact
Train and pair vessel audit	python -m ml.train (seed 2026; 5 x 2 outer folds) python -m ml.refine_vessels
Verify frozen OOF results	python -m ml.evaluate → experiments/metrics.csv
Critical validity checks	python -m pytest
Frontend checks	cd web; npm run typecheck; npm run lint; npm run build
Browser demo reliability	cd web; npx playwright install chromium; npx playwright test
Scientific artifacts	ml/artifacts/{CAD,LAD,LCX,RCA}.joblib; evidence.json; registry.json
Submission materials	README; data/model card; licenses; screenshots; five-minute demo script

Reliability: checkpoint reuse fingerprints data, code, schema, seed, repeats and Python/dependency versions. API verifies all models, OOD, evidence, registry and schema before deserialization. Desktop Chromium integration tests run in CI. Reproduce sensitivity: `python -m ml.reliability`; details: docs/RELIABILITY_AUDIT.md.

Scope and limitations

Small single-center referral cohort with high CAD prevalence; no external clinical validation or longitudinal outcome prediction. Independent classifiers do not produce a validated joint probability. Vessel estimates are weaker, with low recall at the default threshold. Mixed-feature OOD and refit ranges are descriptive heuristics. No diagnosis, treatment advice, stenosis severity or lesion coordinates. Human review is required before submission; the final YouTube recording/upload uses the submitter's account.

Sources, licenses and AI disclosure

Official task: supplied **Track A.pdf**. Clinical data: Alizadehsani, Roshanzamir & Sani, UCI Extension of Z-Alizadeh Sani, **DOI 10.24432/C5461K**, CC BY 4.0. Anatomy: **BodyParts3D / DBCLS**, CC BY 4.0; Mitsuhashi et al., DOI 10.1093/nar/gkn613; derivative pinned to human-atlas commit 1c38bf35c254a891200d3cedecfd57abebe83d8d. Original application code MIT; Manrope and IBM Plex Mono fonts OFL. Detailed attribution and dependency terms: ATTRIBUTIONS.md.

Method references: scikit-learn nested CV and probability calibration documentation; SHAP permutation explanations. Exact URLs and prior-winner research are in docs/RESEARCH.md. The six-page limit applies to this report; full audit tables remain in the software and artifacts.

AI assistance: OpenAI Codex assisted research, coding, debugging, verification and drafting. The deployed product uses no generative model to predict, diagnose or advise. All reported model measurements are computed from the supplied cohort, not generated or copied.